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Hardware Accelerator programming models and their Application for Different computing needs.

A) Hardware accelerator like GPUs, TPUs, FPGAs and ASICs are specialleted devices designed to speed up computation beyond what general-purpose CPUs can offer. To effectively use these accelerators different programming models are adopted each suited to specific workloads.

Programming models such as CUDA and OpenCL enable parallel programming on GPUs, making them ideal for data-intensive tasks like machine learning image processing and scientific simulations. OpenMP and OpenACC provide directive-based model that simplify parallel programming, allowing developers to offload computations to accelerators with the minimal code changes.